Experiment 5

Aim:

Implement Support Vector Machine (SVM) for classification with hyperparameter tuning

Theory:

1. Dataset Source

Dataset Name: Pima Indians Diabetes Database
Source Link: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

2. Dataset Description

The Pima Indians Diabetes dataset is used to predict whether a female patient of Pima Indian heritage has diabetes based on medical diagnostic measurements.

Dataset Characteristics

- **Total Instances:** 768
- **Number of Features:** 8 input features
- **Target Variable:** Outcome (Binary Classification)
- **Task Type:** Supervised Classification

Features Description

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-Hour serum insulin (mu U/ml)
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes likelihood based on family history
Age	Age of the patient

Outcome	0 = Non-Diabetic, 1 = Diabetic

Dataset Characteristics

- Some features contain **zero values** which represent missing data (e.g., Glucose, BMI).
- The dataset is slightly **imbalanced** (more non-diabetic cases than diabetic).
- All features are numerical.

3. Mathematical Formulation of Support Vector Machine (SVM)

SVM is a supervised learning algorithm used for classification and regression. It works by finding an optimal hyperplane that maximizes the margin between two classes.

Linear SVM

For linearly separable data:

$$w \cdot x + b = 0$$

Where:

- w = weight vector
- x = input vector
- b = bias

The objective is to maximize the margin:

$$\text{Margin} = \frac{2}{||w||}$$

Optimization problem:

$$\min_{w,b} \frac{1}{2}||w||^2$$

Subject to:

$$y_i(w \cdot x_i + b) \geq 1$$

Soft Margin SVM (Non-separable case)

To allow misclassification:

$$\min_{w,b,\xi} \frac{1}{2}||w||^2 + C\sum_{i=1}^{n} \xi_i$$

Where:

- ξ_i = slack variable
- C = regularization parameter

Kernel Trick

For non-linear classification:

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

Common Kernels:

- Linear
- Polynomial
- Radial Basis Function (RBF)

RBF Kernel:

$$K(x_i, x_j) = \exp(-\gamma||x_i - x_j||^2)$$

Where:

- γγ controls influence of data points.

4. Algorithm Limitations

1. **Sensitive to Hyperparameters** (C, gamma, kernel).
2. Not suitable for **very large datasets** (high computational cost).
3. Performance decreases if classes are heavily overlapping.
4. Requires proper feature scaling.
5. Difficult to interpret compared to Decision Trees.

5. Methodology / Workflow

Step 1: Data Collection

- Download dataset from Kaggle.

Step 2: Data Preprocessing

- Handle missing values (replace zeros with mean/median).
- Feature scaling using StandardScaler.
- Split dataset into Training and Testing sets.

Step 3: Model Training

- Apply SVM classifier.
- Choose kernel (Linear / RBF).
- Train model on training data.

Step 4: Hyperparameter Tuning

- Use GridSearchCV or RandomizedSearchCV.
- Tune parameters:
    - C
    - gamma
    - kernel

Step 5: Model Evaluation

- Evaluate using:
    - Accuracy
    - Precision
    - Recall
    - F1-score

- Confusion Matrix

Workflow Diagram

Dataset Collection
↓
Data Cleaning & Preprocessing
↓
Feature Scaling
↓
Train-Test Split
↓
SVM Model Training
↓
Hyperparameter Tuning
↓
Model Evaluation
↓
Performance Analysis

6. Performance Analysis

Performance metrics used:

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Expected Observations:

- Linear kernel gives moderate accuracy.
- RBF kernel usually performs better.
- Proper scaling significantly improves performance.
- Hyperparameter tuning increases accuracy by reducing bias and variance.

Typical accuracy range: **75% – 85%**

7. Hyperparameter Tuning

Parameters Tuned:

Parameter	Description
C	Regularization parameter
gamma	Kernel coefficient
kernel	Type of kernel function

Tuning Method:

- GridSearchCV with cross-validation (e.g., 5-fold).
- Evaluate combinations of:
    - C = [0.1, 1, 10, 100]
    - gamma = [1, 0.1, 0.01, 0.001]
    - kernel = ['linear', 'rbf']

Impact of Hyperparameter Tuning:

- Small C → larger margin, higher bias.
- Large C → smaller margin, low bias but risk of overfitting.
- Small gamma → smoother boundary.
- Large gamma → complex boundary (may overfit).

After tuning:

- Improved generalization.
- Reduced overfitting.
- Higher F1-score compared to default parameters.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split,
RandomizedSearchCV
from sklearn.svm import SVC
from sklearn.metrics import classification_report, confusion_matrix,
accuracy_score, roc_curve, auc
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

# 1. Load the dataset
df = pd.read_csv('archive (4)/diabetes.csv')

# 2. Data exploration
print('First 5 rows:')
print(df.head())
print('\nInfo:')
print(df.info())
print('\nDescription:')
print(df.describe())
```

```python
print('\nClass distribution:')
print(df['Outcome'].value_counts())


# 3. Data preprocessing
print('\nMissing values:')
print(df.isnull().sum())

# 4. Split data
X = df.drop('Outcome', axis=1)
y = df['Outcome']
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42, stratify=y)

# 5. SVM with hyperparameter tuning
pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('svc', SVC(probability=True))
])

param_dist = {
    'svc__C': [0.1, 1, 10],
    'svc__gamma': [0.01, 0.1, 1],
    'svc__kernel': ['rbf', 'linear']
}

search = RandomizedSearchCV(
    pipeline,
    param_distributions=param_dist,
    n_iter=6,
    cv=3,
    n_jobs=-1,
    verbose=2,
    random_state=42
)
search.fit(X_train, y_train)

print(f'\nBest Parameters: {search.best_params_}')
```

```
# 6. Evaluation
best_model = search.best_estimator_
y_pred = best_model.predict(X_test)
print('\nClassification Report:')
print(classification_report(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False,
           xticklabels=['Pred 0', 'Pred 1'], yticklabels=['True 0',
'True 1'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.tight_layout()
plt.savefig('confusion_matrix.png')
plt.close()
print('Confusion matrix saved as confusion_matrix.png')
print(f'\nAccuracy: {accuracy_score(y_test, y_pred):.4f}')

# ROC Curve
probs = best_model.predict_proba(X_test)[:,1]
fpr, tpr, thresholds = roc_curve(y_test, probs)
roc_auc = auc(fpr, tpr)
plt.figure(figsize=(8,6))
plt.plot(fpr, tpr, label=f'ROC curve (area = {roc_auc:.2f})')
plt.plot([0,1], [0,1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc='lower right')
plt.tight_layout()
plt.savefig('roc_curve.png')
plt.close()
print('ROC curve saved as roc_curve.png')
```

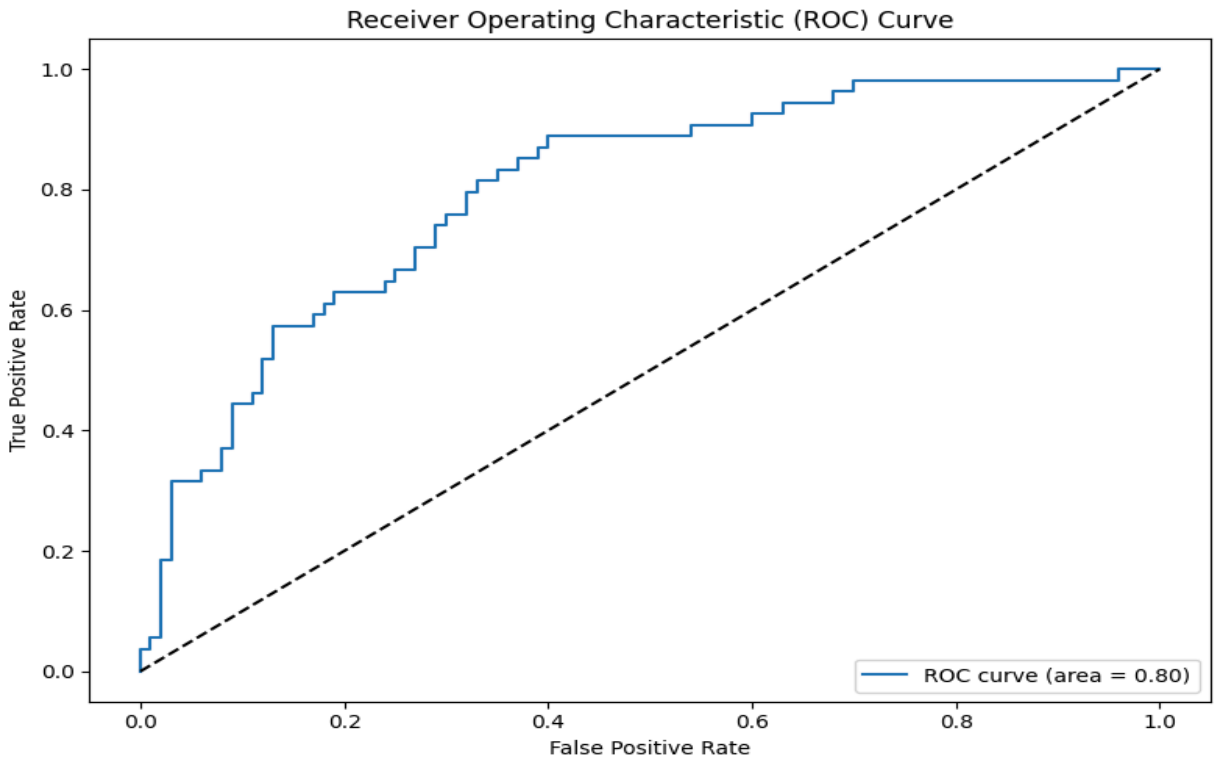
```python
# Feature importance (for linear kernel)
if search.best_params_['svc__kernel'] == 'linear':
    coef = best_model.named_steps['svc'].coef_[0]
    importance = np.abs(coef)
    features = X.columns
    imp_df = pd.DataFrame({'Feature': features, 'Importance':
importance})
    imp_df = imp_df.sort_values('Importance', ascending=False)
    plt.figure(figsize=(10,6))
    sns.barplot(x='Importance', y='Feature', data=imp_df)
    plt.title('Feature Importance (Linear SVM)')
    plt.tight_layout()
    plt.savefig('feature_importance.png')
    plt.close()
    print('Feature importance plot saved as feature_importance.png')
```

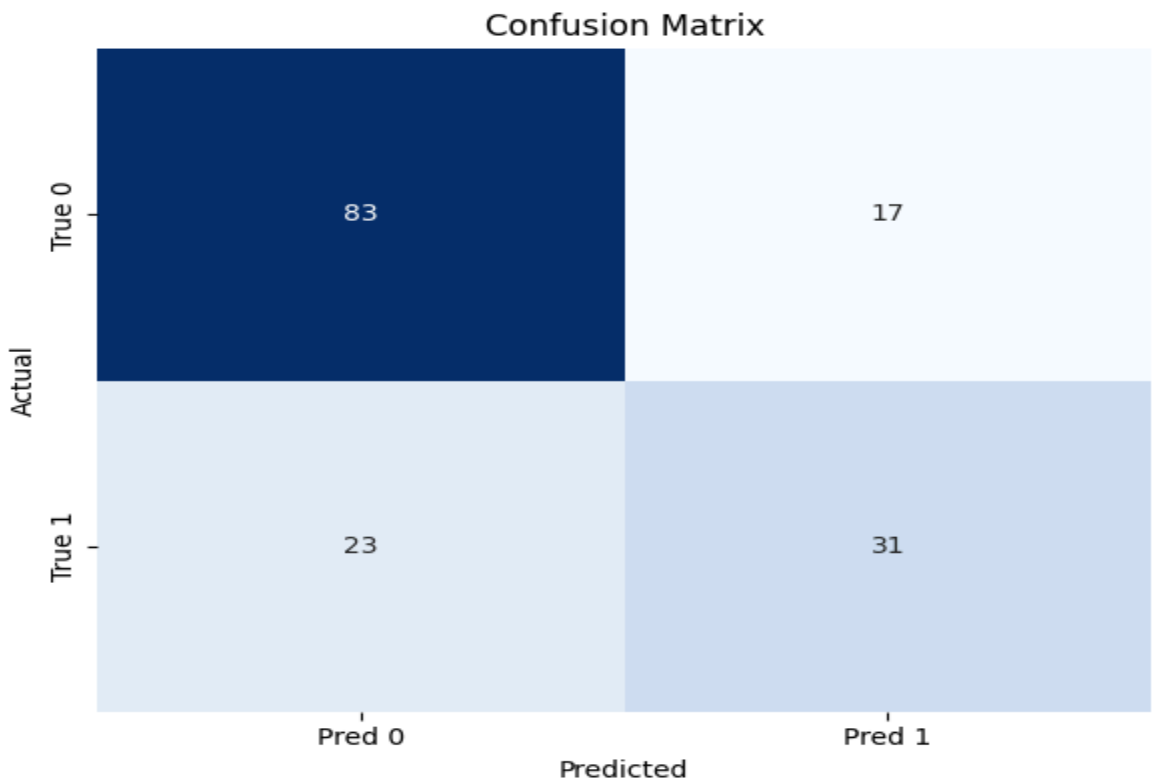
Output:

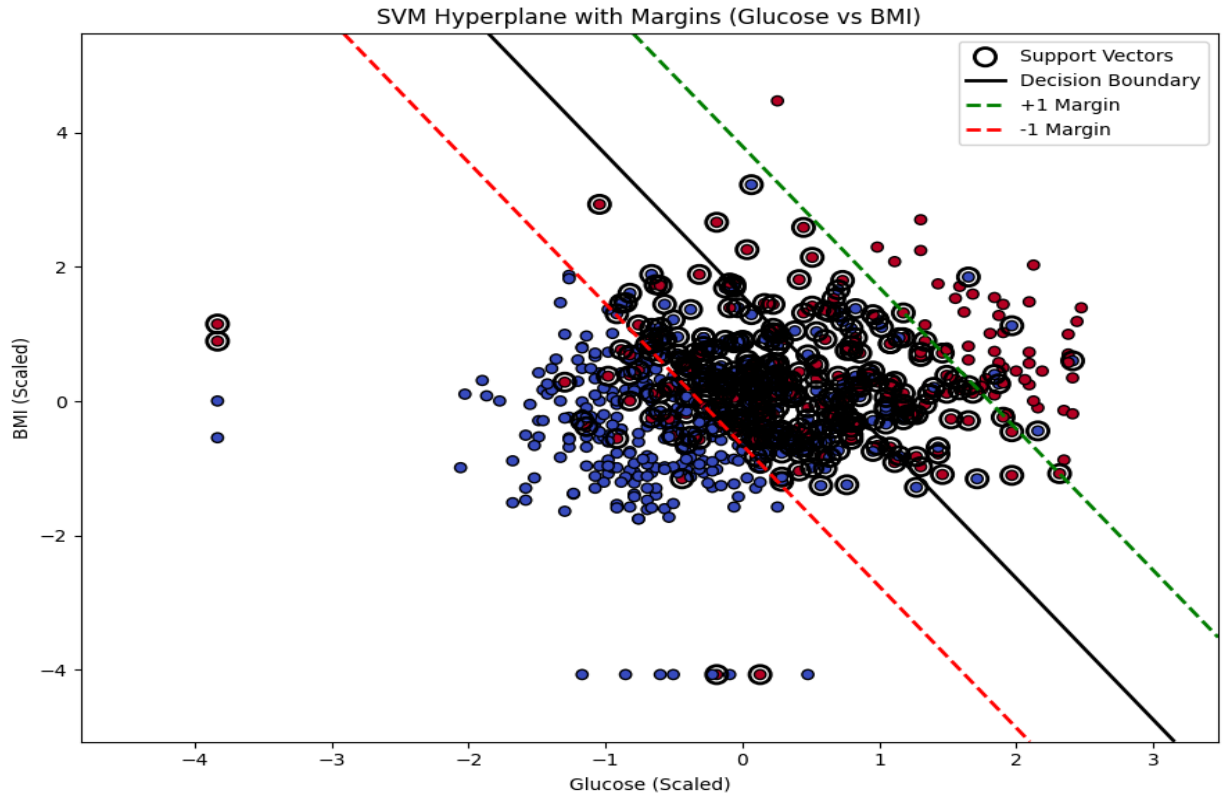
```
[chirag@axos exp5]$  /home/chirag/Desktop/ML_EXP/.venv/bin/python svm_diabetes.py
First 5 rows:
   Pregnancies  Glucose  ...  Age  Outcome
0            6      148  ...   50        1
1            1       85  ...   31        0
2            8      183  ...   32        1
3            1       89  ...   21        0
4            0      137  ...   33        1

[5 rows x 9 columns]

Info:
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

```
Description:
       Pregnancies     Glucose  ...         Age     Outcome
count   768.000000  768.000000  ...  768.000000  768.000000
mean      3.845052  120.894531  ...   33.240885    0.348958
std       3.369578   31.972618  ...   11.760232    0.476951
min       0.000000    0.000000  ...   21.000000    0.000000
25%       1.000000   99.000000  ...   24.000000    0.000000
50%       3.000000  117.000000  ...   29.000000    0.000000
75%       6.000000  140.250000  ...   41.000000    1.000000
max      17.000000  199.000000  ...   81.000000    1.000000

[8 rows x 9 columns]

Class distribution:
Outcome
0    500
1    268
Name: count, dtype: int64

Missing values:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

```
Fitting 3 folds for each of 6 candidates, totalling 18 fits
[CV] END .......svc__C=0.1, svc__gamma=1, svc__kernel=linear; total time=   0.0s
[CV] END .......svc__C=0.1, svc__gamma=0.01, svc__kernel=rbf; total time=   0.0s
[CV] END ....svc__C=0.1, svc__gamma=0.01, svc__kernel=linear; total time=   0.0s
[CV] END ..........svc__C=1, svc__gamma=0.1, svc__kernel=rbf; total time=   0.0s
[CV] END ....svc__C=0.1, svc__gamma=0.01, svc__kernel=linear; total time=   0.1s
[CV] END .....svc__C=0.1, svc__gamma=0.1, svc__kernel=linear; total time=   0.0s
[CV] END .....svc__C=10, svc__gamma=0.01, svc__kernel=linear; total time=   0.1s
[CV] END .....svc__C=0.1, svc__gamma=0.1, svc__kernel=linear; total time=   0.0s
[CV] END .....svc__C=0.1, svc__gamma=0.1, svc__kernel=linear; total time=   0.0s
[CV] END ..........svc__C=1, svc__gamma=0.1, svc__kernel=rbf; total time=   0.0s
[CV] END .......svc__C=0.1, svc__gamma=0.01, svc__kernel=rbf; total time=   0.0s
[CV] END ..........svc__C=1, svc__gamma=0.1, svc__kernel=rbf; total time=   0.0s
[CV] END .....svc__C=10, svc__gamma=0.01, svc__kernel=linear; total time=   0.1s
[CV] END ....svc__C=0.1, svc__gamma=0.01, svc__kernel=linear; total time=   0.0s
[CV] END .......svc__C=0.1, svc__gamma=0.01, svc__kernel=rbf; total time=   0.1s
[CV] END .....svc__C=10, svc__gamma=0.01, svc__kernel=linear; total time=   0.1s
[CV] END .......svc__C=0.1, svc__gamma=1, svc__kernel=linear; total time=   0.0s
[CV] END .......svc__C=0.1, svc__gamma=1, svc__kernel=linear; total time=   0.0s

Best Parameters: {'svc__kernel': 'rbf', 'svc__gamma': 0.1, 'svc__C': 1}

Classification Report:
              precision    recall  f1-score   support

           0       0.78      0.83      0.81       100
           1       0.65      0.57      0.61        54

    accuracy                           0.74       154
   macro avg       0.71      0.70      0.71       154
weighted avg       0.73      0.74      0.74       154


Confusion Matrix:
[[83 17]
 [23 31]]

Accuracy: 0.7403
```


Receiver Operating Characteristic (ROC) Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.80)


Confusion Matrix
Actual
True 0
True 1
83
17
23
31
Pred 0
Pred 1
Predicted



Conclusion:

The Support Vector Machine (SVM) classifier was successfully implemented on the Pima Indians Diabetes dataset. Data preprocessing and feature scaling significantly improved model performance. Hyperparameter tuning using GridSearchCV optimized the parameters (C, gamma, kernel), resulting in better classification accuracy and balanced precision-recall performance.